



ESPE

UNIVERSIDAD DE LAS FUERZAS ARMADAS
INNOVACIÓN PARA LA EXCELENCIA

DCCO



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Homework 06 Fundamentals OOP

OOP - 23217

Professor: Edison Lascano

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Object: InventoryReportSystem

Class: ReportGenerator

Attributes:

reportStartDate

reportEndDate

reportType

reportData

Methods:

generateReport()

exportToPDF()

sendReportByEmail()

filterByCategory()

Inventario
-reportStartDate: String -reportEndDate: String -reportType: String -reportData: String
+generateReport(): String +exportToPDF(): boolean +sendReportByEmail(email: String): boolean +filterByCategory(category: String): String

Object: Log

Class: Sale

Attributes:

id

date

total

customer

~~salesman~~

Methods:

calculateTotal()

saleRegister()

Sale
id: int date: String total: float

calculateTotal(): return float saleRegister(): return String

Object: Report
Class: ContableReport
Attributes:
id
date
description
~~balance~~
Methods:
generarReport()
export()
deleteReport()

ContableReport
-id: int -date: int -description: String
+generarReport(): return String +export(): return void +deleteReport(): return void

Object: Report
Class: Notification
Attributes:
id
date
message
productId

Methods:
sendNotification()
stockVerify()

Notification
-id: int -date: int -message: String -productId: int
+sendNotification(): return void +stockVerify(): return boolean

Object: sale

Class:

Attributes:

Methods:

id

Venta: String

fecha: Date

|productos: List

total: double

metodo Pago: String

+calcularTotal(): double +generarFactura(): String
+aplicarDescuento(codigo: String): boolean +agregarProducto(producto: Producto, cantidad: int): void

Object:Woman Clothes

Class:

Attributes:

Methods:

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tipoEscote: String longitudFalda: String verificarTallasDisponibles(): String[] sugerirAccesorios(): String[]
esEstampado: boolean

Object: Report

Class: MounthlyContableReport

Attributes:

id

month

incomeTotal

expensesTotal

description

~~generateDate~~

Methods:

totalCalculate()

generateReport()

export()

~~show()~~

MounthlyContableReport
-id: int -month: int -incomeTotal: float -expensesTotal: float -description: String
+totalCalculate(): return void +generateReport(): return void +export(): return void

Object: UserRegistrationSystem

Class: UserManager

Attributes:

userName

userEmail

userRole
registrationDate

Methods:

registerUser()
assignRole()
validateCredentials()
listRegisteredUsers()

UserManager
-userName: String -userEmail: String -userRole: String -registrationDate: String
+registerUser: boolean +assignRole:String +validateCredentials:String +listRegisteredUsers(): String

Object: UserRegistrationSystem

Class: UserManager

Attributes:

userName
userEmail
userRole
registrationDate

Methods:

registerUser()
assignRole()
validateCredentials()
listRegisteredUsers()

HistoryModule
-productId: String -historyRecords: String -dateRange: String -movementType: String
+viewTransactionHistory: String +filterHistoryByDate: String +exportHistoryToExcel: boolean +searchByProduct:String

Object: InventoryUpdater

Class: InventoryManager

Attributes:

productId

quantity

transactionType

updatedStock

Methods:

updateInventory()

checkProductExists()

logInventoryChange()

confirmStockUpdate()

InventoryManager
-productId: String -quantity: int -transactionType: String -updatedStock: int
+updateInventory: String,int, boolean +checkProductExists: String +logInventoryChange: String,int +confirmStockUpdate: boolean

Object: Clothing Supplier

Class: Supplier

Attributes:

id

companyName

contactInfo

suppliedProducts

address

Methods:

registerProduct(product) updateContact(newContact)
viewOrderHistory(): return void deleteSupplier(): return void

Object: System User

Class: Product

Attributes:

id

name

size

color

price

stock_quantity

Methods:

Product
increaseStock(quantity) decreaseStock(quantity)
applyDiscount(percentage) displayDetails()